Hydraulic System Pressure Relief Valve – Maintenance Practices

1. Adjustment/Test

Equipment required:

- Hydraulic test rig using hydraulic fluid (MIL–H–5606) suitably fitted with electric motor-driven hydraulic pump
- Shut-off and metering valves
- Gauge
- Flowmeter
- Graduated cylinder which, in conjunction with the unit being tested, will reproduce all conditions required during the test

A. Bench Test Airframe Hydraulic System Pressure Relief Valve

1) Relieve airframe hydraulic system pressure, by cycling flaps.

2) Remove hydraulic system relief valve from power pack.

3) Install airframe hydraulic relief valve in test rig.

4) Apply 1950 ± 25 psi hydraulic pressure to valve inlet and check that rated flow through valve is 2.10 Imp (2.5 U.S.) gal/min.

5) Reduce inlet pressure to 1680 psi. Check that leakage through valve does not exceed 4.5 cc/min.

Note Leakage to be checked during third minute of a three minute waiting period.

6) Slowly reduce inlet pressure. At pressures of 1460 psi and 975 psi, check that leakage rate through valve does not exceed 1.5 cc/min. Refer to Note in Para 1.A.5).

7) Slowly increase inlet pressure to 1460 psi. Repeat leakage check given in Para 1.A.6).

8) Increase inlet pressure to 1680 psi. Repeat leakage check given in Para 1.A.5).

9) Reduce inlet pressure to zero and remove valve from test rig. If satisfactory and required for service, install in aircraft.

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